

Terrain-Interaction Control of a Hexapod Robot

A measured comparison of three controllers on a shared simulation plant

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C1

C2

C3

The question

*Terrain search buys terrain adaptation without adding a single sensor.
What does it cost where there is no terrain to discover?*

Falsifiable

The reference work shows the mechanism helps on unmodelled terrain. It never reports its price on terrain the gait already models.

Isolable

Three controllers, one plant. Each adds exactly one mechanism, so every adjacent comparison has one cause.

Declarable

A measured noise floor decides which differences are results and which are not.

The platform and the reference work

- PhantomX AX Metal Hexapod Mark II
- 18 Dynamixel AX-12A servos, three per leg
- No foot force sensors, no IMU
- The only feedback is each servo's own position and load

Arrigoni et al., Robotics 2024

Detect foot-ground contact from motor torque alone: compare the measured joint torque against an inverse-dynamics prediction, and use the detection to adapt commanded foot height.

$$\text{contact} = \text{OR over } j \left(\left| \text{tau_meas} - \text{tau_exp} \right| > \text{sigma_j} \right)$$

Why we picked it

01

The claim has a price nobody quoted

A controlled comparison on identical ground can measure it. The answer turned out to be a real and systematic cost.

02

The contact detector is a model-based observer in disguise

Its quality depends on the accuracy of a dynamic model. We found out the hard way how directly.

03

A shared plant makes the ablation clean

Same ground, same contact law, same gait generator. Rarely available when comparing published controllers.

Three controllers, one mechanism apart

C1

Open-loop kinematic tripod

Foot trajectories from the gait parameters, inverse kinematics per leg, joints in position. No feedback.

C2

C1 + terrain search

While a leg is in stance and the contact flag is low, commanded foot height keeps descending until the torque residual crosses threshold.

C3

C2 + attitude loop

Two discrete PI regulators on pitch and roll; their output is distributed over the six feet as an additive height offset.

terrain_search → z_i → $\Sigma (+\Delta z_i)$ → Saturation → Rate limiter → leg

Verified by reading block connectivity inside the model file, not from documentation. C3 contains C2: C2→C3 isolates the attitude loop, C1→C2 the search.

01

The test bench

Imported from the public PhantomX URDF into Simscape Multibody

The plant

- Imported from the public PhantomX URDF into Simscape Multibody
- 25 rigid bodies, 18 revolute joints, 6 spherical contact elements
- Joints commanded in position: torque is a reaction, not an input
- Penalty contact law with Coulomb friction

4.3 mm

static penetration in tripod stance, against a 2.0 mm transition region — tuned so the force law stays well defined

[FIGURE]

Simscape model / plant block diagram

Gait parameters and where they come from

The reference work delegates the gait to NUKE and declines to discuss it. Three of five parameters are fixed by a measured constraint, and only one degree of freedom is left.

$$T = S / (\text{beta_st} \cdot v_nom)$$

phase vector	—	Definition of the alternating tripod. Not a tunable.
beta_stance	0.50	The only duty factor giving continuous support on exactly three legs.
S	0.06 m	80.9% of leg reach; the internal assert trips at 85%, i.e. S = 0.095 m.
H	0.05 m	1.5x the measured obstacle height, 32.8 mm, read from the stance-foot elevation.
T	1.00 s	Dependent variable. Gives v_nom = 0.12 m/s, at 83% of the measured envelope.

Froude number $v^2/(gh) = 0.0095$ — inertial forces are one percent of gravity, so the march is firmly quasi-static.

Seven tasks

T2	Flat, speed sweep	Baseline and the stability envelope in speed
T3	Curved path	Yaw command 0.1 rad/s
T4	8 deg ramp	Continuous slope: body attitude versus terrain
T4D	8/8 deg bump	Up then down, two angular interfaces
T5	Single obstacle	One unmodelled step, 32.8 mm
T6	Seven obstacles	Repeated discontinuities, left and right raised alternately
T7	Lateral impulse	Momentum step from 1% to 1600% of nominal

[FIGURE]

Terrain configurations: flat, ramp, bump, obstacle, course

02

Method

A difference is declared only if it is at least three times the worse of the two noise floors.

When is a difference a result?

Every run is repeated three times with the stance height displaced by ± 0.2 mm — a quantity that has nothing to do with the controller. The resulting excursion is that metric's noise floor.

A difference is declared only if it is at least three times the worse of the two noise floors.

Not a formality

Applying it removed several differences that looked like results. Roughly a third of what we measured does not survive.

Including our own

Every peak-torque comparison fails the rule, on every task — despite differences as large as +87%.

Two more rules, each adopted after breaking it

The criterion is written before the run

Every diagnostic script carries, in its header, the question it asks, the threshold it will apply and the possible conclusions — all written before the data exist. A threshold calibrated afterwards would have moved the stability limit from 1.0x to 1.2x depending on the multiple we chose.

Retractions stay written

Every withdrawn claim is marked in place, with the reason. We withdrew at least five. One wrong claim survived three days precisely because it was plausible.

03

Two defects

For three days C2 did not search the terrain. Nothing signalled it: the robot walked, the animation was plausible, the metrics came out.

URDF inertias: wrong by a factor of about one thousand

Radius of gyration

From $I = m r^2$, the declared trunk inertia places the trunk's mass 1.79 m from its own centre. The trunk is 25 cm across. Every link gives 0.46 m; the links are 2 to 12 cm long.

All 24 links identical

Same inertia to the same decimal, for coxa, femur and tibia. A placeholder, not a measurement.

Dividing by 10^3 brings everything into the physically plausible range — the signature of a CAD export with masses in grams.

$45^\circ \rightarrow 15^\circ$

peak pitch of C1 on rough ground

$1.35 \rightarrow 0.68$

cost ratio C2/C1 — it changed sign

Correcting the robot broke the contact detector

We corrected the inertias of the simulated robot only. From that moment the robot and the model predicting its torque were two different robots, three orders of magnitude apart.

The residual became large always, with or without contact. The flag stayed high over 95% of swing, so the terrain search never entered the branch that grows foot extension — and the command degenerated into a constant.

For three days C2 did not search the terrain. Nothing signalled it: the robot walked, the animation was plausible, the metrics came out.

	misaligned	aligned
Flag error during stance	7.0%	0.6%
Error of an always-high flag	7.0%	3.2%
Does the flag carry information?	no	yes
Swings with no reset of the search	20/60	2/60

> 95%

of the swing phase with the contact flag stuck high

Two corrections that are our own choice

Contact threshold re-tuned

At the original threshold the flag still stayed high over 41.9% of swing, and 39.7% of that was the coxa alone: it rotates about the vertical axis while contact is vertical, so contact transmits almost no torque to it. The adopted triple switches the coxa off — a change of rule, not a re-tuning: the flag becomes an OR over two joints.

Slew-rate limiter in both plants

The attitude model carried a rate limiter the open-loop plant did not have — an asymmetry that would have confounded the attitude loop with its own limiter. We placed ± 0.4 m/s in both. The nominal gait peaks at 0.375 m/s, so the nominal march is untouched.

The limiter mattered more than expected: it removed C2's lateral drift on the obstacle course, from 0.296 to 0.104 m. An implementation detail, with effects larger than some of the differences between controllers.

04

Results

On ground that is exactly what the trajectory generator believes it to be, terrain search makes attitude three to four times worse and buys nothing. The mechanism keeps working where there is nothing to find, and the body pays for it.

Flat and curve: terrain search costs attitude

	C1	C2	C3	C1→C2	C2→C3
T2 pitch max [deg]	0.514	1.95	0.356	+279% (16.3x)	-82% (18.1x)
T2 cost of transport	0.888	0.865	1.11	-3% (n.c.)	+28% (27.7x)
T3 pitch max [deg]	0.579	1.85	0.392	+219% (40.0x)	-79% (45.9x)
T3 roll max [deg]	0.224	0.351	0.210	+57% (4.6x)	-40% (5.1x)
T3 energy [J]	30.3	29.5	37.8	-3% (6.4x)	+28% (22.5x)

On ground that is exactly what the trajectory generator believes it to be, terrain search makes attitude three to four times worse and buys nothing. The mechanism keeps working where there is nothing to find, and the body pays for it.

[FIGURE]

*Body pitch over two gait cycles, flat ground,
three controllers*

Discontinuities: terrain search repays

	C1	C2	C3	C1→C2	C2→C3
T4D pitch max [deg]	8.65	10.4	1.98	+21% (19.8x)	-81% (93.8x)
T4D roll max [deg]	2.12	1.49	0.611	-30% (6.4x)	-59% (7.6x)
T4D energy [J]	64.1	59.9	83.6	-7% (3.5x)	+40% (18.8x)
T5 pitch max [deg]	7.56	6.41	1.73	-15% (16.4x)	-73% (19.8x)
T5 cost of transport	1.20	1.08	1.40	-10% (5.2x)	+30% (17.1x)

Here the picture reverses: the search improves roll, energy, cost of transport and speed. One solid exception on the bump — peak pitch is 21% worse, at 19.8 times the noise floor.

[FIGURE]

Body attitude during the obstacle crossing, with the crossing window shaded

The ramp: what the attitude loop is for

	C1	C2	C3
Body attitude minus ramp angle [deg]	0.106	0.259	-7.54
Peak-to-peak pitch on the ramp [deg]	0.994	1.97	0.839
Roll on the ramp [deg]	0.662	0.563	0.314
Cost of transport	1.15	1.09	1.58

Noise floor not measured on this task: no verdicts are issued.

C1 and C2 align the body with the ramp. C3 holds it within half a degree of horizontal — two different behaviours, not two qualities of the same one, which is why that row carries no percentage verdict. It costs 45% in cost of transport.

[FIGURE]

Body pitch while climbing the 8 deg ramp, with the slope as reference

Obstacle course: the comparison does not discriminate

	C1	C2	C3
Distance travelled [m]	2.91	3.12	2.99
Max lateral drift [m]	0.067	0.104	0.058
Course completed, valid	no	no	no

"Completed" counts only below 0.183 m of lateral drift, a threshold from the obstacle geometry, computed independently of any controller. Beyond it the robot may have missed an obstacle entirely rather than stepped over it.

None of the three completes the course. The fine metrics must not be quoted: the three follow different trajectories and do not meet the same obstacles. On this task the comparison does not separate the controllers — and saying so is the result.

Lateral impulse: a gap common to all three

	C1	C2	C3
Residual deviation, 1x [mm]	1.34	6.04	0.164
Residual deviation, 16x [mm]	393	400	392
Recovers at 16x	no	no	no

Noise floor not measured on this task: the 97% improvement at 1x is reported without a verdict.

0.164 mm

residual deviation of C3 at 1x impulse,
against 1.34 mm for C1

2%

spread between the three at 16x: none
of them recovers

At large impulse the three coincide to within 2%. Contact detection is not the mechanism in play: this is an architectural gap shared by all three, and the clearest motivation for any future controller on this platform.

Feasibility on the real servos

The campaign runs with ideal actuators. Since joints are commanded in position, what follows measures what each controller demands, not how it would behave with real motors.

39–46%

of stall on the worst-loaded joint, every task,
every controller

2.3x

the continuous load ROBOTIS recommends for
stable motion

$\leq 0.91\%$

of samples above stall: isolated impact peaks
reaching 5.0x

[FIGURE]

Commanded torque against the AX-12A limit: continuous and peak, per task

C3 consistently demands the lowest continuous torque and the lowest impact peaks on the tasks with discontinuities: the attitude loop absorbs part of the impact in body motion rather than in the joints.

05

What it means

Composed in series, C3 ends up better than C1 on pitch: the attitude loop more than pays back the disturbance the search introduced.

The two mechanisms do different things

Terrain search

C1 → C2

- Not an attitude mechanism: pitch +219 to +279° where the ground is already modelled
- On real terrain: roll -30 to -49°, cost of transport -7 to -10%, speed +2 to +3%
- Pitch still worsens by 21° on the bump
- Why: contact is declared per leg, asynchronously, and the height correction is a step

Attitude feedback

C2 → C3

- Buys attitude on every task: pitch -73 to -82°, at 18 to 94 times the noise floor
- On the ramp, a qualitatively different behaviour
- Costs 28 to 40% in energy and cost of transport
- Spent on foot travel, not joint load: continuous torque actually falls

Composed in series, C3 ends up better than C1 on pitch: the attitude loop more than pays back the disturbance the search introduced.

Unplanned finding: stance becomes unobservable

On the ramp, the kinematic stance detector found almost no stance on C3. Our first reading was that the robot was not walking. It was wrong — the mean body height we quoted was the ramp's own height, and the force sensors see only the floor plane.

	samples under 0.05 m/s	longest consecutive stretch
C2	41-54%	71-113 samples (one full stance)
C3	4-8%	3-12 samples

The feet are not failing to land: they land and are moved again. The attitude loop adds a continuous height correction to every foot, stance feet included, so a foot in stance is never at rest in the world frame. The amplitude is small — about 17 mm — so the gait is unaffected.

What breaks is the observability of contact by any rest-based criterion. Composing an attitude loop after a contact-triggered mechanism feeds the second one a world in which nothing is ever stationary.

Limitations

Ideal actuators

Torque is demand, not achievable behaviour. Servo limits in the loop would likely compress the differences.

Sensors see only the floor

On ramp, bump and obstacles every stance metric comes from the kinematic detector — itself degraded on C3.

Lever arms hardcoded

About 1.8x smaller than nominal foot positions. Not corrected: the gains were tuned with those values in the loop.

Self-tuned threshold

Fitted on this bench, so the search cannot be credited with robustness to it.

Estimator is not the plant

Residual mismatch is the 1.6% of mass in the feet, which the URDF tree does not carry.

No hardware

Nothing here was measured on a physical PhantomX. The comparison is not validated.

Contributions

1

[Student 1]

Plant and controllers

- Simscape Multibody plant from the URDF
- Open-loop tripod generator (C1)
- Terrain-search layer, residual and contact flag (C2)
- Attitude loop: design, implementation, empirical tuning (C3)
- Saturation and rate limiter

2

[Student 2]

Measurement and analysis

- Task battery and measurement protocol
- Noise floor and the 3x declarability rule
- Campaign and its automation
- The two bench defects: found, measured, corrected
- Stance diagnosis, stability check, feasibility

Joint: choice of the three controllers and of the ablation structure, gait provenance, interpretation of the results, report and deck.

Conclusions

- Terrain search alone is a net cost on terrain the gait generator already models: pitch +219 to +279% with no energy return. On irregular terrain it improves roll, cost of transport and speed, while still worsening pitch.
- The attitude loop is the mechanism that buys attitude, on every task, at 18 to 94 times the noise floor. It costs 28 to 40% in energy, spent on foot travel rather than joint load.
- Composed in series the two leave the body better than open loop, but make foot contact unobservable to any rest-based detector.
- On the real servos the march itself would run at 2.3x the recommended continuous load.

What comes next

- The fourth cell of the ablation: attitude feedback without terrain search
- A campaign with the servo limits in the loop, where C3's energy cost stops being free

C1

C2

C3